CSE 221 Summer 24 Quiz 01

SET A

TIME: 30 Minutes

Name	ID	Section

1.	Given the time function T(n) [which is the actual function], and another function g(n)[possible asymptotic Notation], determine if t(n) can be projected/bound as O(g(n), Ω(g(n) and θ(g(n))? You can use the proof technique for upper bound and lower bound notation to justify the answer. • $T(n) = \sqrt{n} + log_2^n,\ g(n) = n^{0.5}$ • $T(n) = nlogn,$ g(n) = $log(n^2)$	**3+3**
2.	Consider the code Snippet and identify the time complexity of the program. O(n) notation will be fine. code1: **x = 1** **y = n** **while (y<1):** **y = y - x** **x += 1** code2: **x=100** **for h in range(x*x):** **print(" :(")**	**4+2**
3.	Solve the time complexity of the recursive function: [Any method is acceptable if you do it correctly with a proper explanation not just a plain answer] $T(n) = T(n/2) + T(n/7) + n$	**8**

CSE 221 Summer 24 Quiz 01

SET B

TIME: 30 Minutes

Name	ID	Section

1.	Given the time function T(n) [which is the actual function], and another function g(n)[possible asymptotic Notation], determine if t(n) can be projected/bound as O(g(n), Ω(g(n) and θ(g(n))? You can use the proof technique for upper bound and lower bound notation to justify the answer. • $T(n) = 10^8 n + n^2$, $g(n) = n^{1.99}$ • $T(n) = 2.33^n + log_2^n$, $g(n) = 3^n$	**3+3**
2.	Consider the code Snippet and identify the time complexity of the 2 programs. O(n) notation will be fine. **code1:** **c = 0** **x=30** **while c<n:** **x= x + c** **c += 2** **Code2:** **x=n** **s=13** **while (x>0):** **s*=s*(12**x)** **print("algo")** **x-=x**	**4+2**
3.	Solve the time complexity of the recursive function: [Any method is acceptable if you do it correctly with a proper explanation not just a plain answer] $T(n) = T(n-1) + T(n-3) + 1$	**8**

CSE 221 Summer 24 Quiz 01

SET C

TIME: 30 Minutes

Name	ID	Section

1.	Given the time function T(n) [which is the actual function], and another function g(n)[possible asymptotic Notation], determine if t(n) can be projected/bound as O(g(n), Ω(g(n) and θ(g(n))? You can use the proof technique for upper bound and lower bound notation to justify the answer. • $T(n) = n^{0.1}$, $g(n) = (logn)^{10}$ • $T(n) = 10^{18} + log_2^n$, $g(n) = log_3^n$	**3+3**
2.	Consider the code Snippet and identify the time complexity of the 2 programs. O(n) notation will be fine. code1: **add=1** **for x in range(2n,0,-3):** **for y in range(x,0,-1):** **for c in range(0,x*x,2):** **add = add/(c//x)** **Break** Code2: **x=n** **s=13** **while (x>0):** **s*=s*(12**x)** **print("algo")** **x-=x**	**4+2**
3.	Solve the time complexity of the recursive function: [Any method is acceptable if you do it correctly with a proper explanation not just a plain answer] $T(n) = T(n/2) + T(n/3) + 1$	**8**

CSE 221 Summer 24 Quiz 01

SET D

TIME: 30 Minutes

Name	ID	Section

1.	You are given 2 functions and their upper bound asymptotic notations. Define the properties of the upper bound functions in the format: T(n) = O(g(n)) ---> [0 <= T(n)<= c*g(n)] for all n > n0. **T(n)= n2+ 100: O(n2); what will be the value of n0 if c = 2?** **T(n)= n + 10: O(n); what will be the value of n0 if c = 3?**	**2+2**

2.	Consider the code Snippet and identify the time complexity of the 2 programs. O(n) notation will be fine.
	
	code1:
	x=100
	for h in range(x*x):
	print(" :(")
	
	Code2:
	y = n
	while(y>0):
	y=y//2
	print("running")
		2+4

3.	$T(n) = (n)!$ Is it valid to write this function as O(n^n); Ω(n^n) and θ(n^n)? Just write Yes/No for each case.	**3**
4.	Solve the time complexity of the recursive function: [Any method is acceptable if you do it correctly with a proper explanation not just a plain answer] $T(n) = T(n/4) + T(n/8) + 1$	**7**

CSE 221 Summer 24 Quiz 02
SET A
TIME: 30 Minutes

Name	ID	Section

1.	Your friend gave you a binary string B (meaning each character is either 0 or 1). He wanted to find out how to calculate the maximum number of consecutive 0s in that particular string. For example, String: **100100000111 Maximum consecutive 0s: 5** String: **1010101010101 Maximum consecutive 0s: 1** You, as an algorithm enthusiast, know that this can be solved in linear time. However, your friend asked you to propose a divide-and-conquer approach. **A) Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.** **B) Write the time complexity of your algorithm.**	**10**
2.	**Write a simulation for mergesort when its the worst-case scenario[Array of length 7]. You have to properly show the simulation with each step.**	**10**

CSE 221 Summer 24 Quiz 02

SET B

TIME: 30 Minutes

Name	ID	Section

1.	Your friend gave you an encoded message where each number can only be either a multiple of 2 or not. He wanted to find out how to calculate the maximum number of consecutive multiples of 2s in that particular string. For example, String: **122124628111 Maximum consecutive multiple of 2s: 5** String: **2121010210221 Maximum consecutive multiple of 2s: 2** You, as an algorithm enthusiast, know that this can be solved in linear time. However, your friend asked you to propose a divide-and-conquer approach. **A) Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.** **B) Write the time complexity of your algorithm.**	**10**
2.	**Write a simulation for mergesort when its the best-case scenario[Array of length 7]. You have to properly show the simulation with each step.**	**10**

CSE 221 Summer 24 Quiz 02

SET C

TIME: 30 Minutes

Name	ID	Section

1.	**Given a binary string S of type{1^m0^n} exactly in this order.some 1s followed by some 0s. No additional 1 can be added at the end of the string, devise an algorithm that finds the number of zeros in O(log k) time. Let m + n = k** **Example:** **111100000000, Answer: 8** **111111000, Answer: 3** **1111111111, Answer: 0** **Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.**	**8**
2.	**Write a simulation for Quick Sort when its the worst-case scenario[Array of length 6]. You have to properly show the simulation with each step.**	**12**

CSE 221 Summer 24 Quiz 02

SET D

TIME: 30 Minutes

Name	ID	Section

1.	**Given a string S of type{X^mY^n} exactly in this order.some Xs followed by some Ys. No additional X can be added at the end of the string, devise an algorithm that finds the number of X and Y 1s in O(log k) time. Let m + n = k** **Example:** **XXXXXYYY, Answer: X: 5 and Y: 3** **XYY, Answer: X: 1 and Y: 2** **YYYYY, Answer: X: 0 and Y: 5** **Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.**	**8**
2.	**Write a simulation for Quicksort when its the best-case scenario[Array of length 8]. You have to properly show the simulation with each step.**	**12**